

Ans: B

Q7.

From graph, $F = kx$ or $4.5 = k (0.200 - 0.050) \Rightarrow k = 30.0 \text{ N m}^{-1}$

By the Law of Energy Conservation, Loss in GPE of load = Gain in elastic PE of spring

$$mgx = (1/2)kx^2 \text{ or } 0.150 \times 9.81 (x) = (1/2) (30.0) x^2$$

$$x (15.0 x - 1.47) = 0 \text{ or } x = 0.0981 \text{ m} = 9.81 \text{ cm}$$

So, maximum length attained = $5.0 + 9.81 = 14.8 \text{ cm}$

Ans: C

Q8.